Follow-up experiments: extra seeds, ℓ_min = 1 and validation-selected thresholds

This report covers the three parts of the follow-up to the Protocol-A ablation study of the HybridSBD shot boundary detection models (leak-free SHOT training set of 448 videos): **Part 1** trains A0 and A1 with two more seeds; **Part 2** re-evaluates every model with ℓ_min = 1 on all three test sets (Protocol A with ℓ_min = 1); **Part 3** evaluates every model with thresholds selected on the SHOT validation split (Protocol B). A short additional check (validation F1 per epoch) follows. Evaluation uses the existing best checkpoints (EMA weights selected by SHOT validation F1). Windows (T = 128, stride 64), window averaging, threshold grids (τ_c 0.10–0.90 step 0.05, τ_g ∈ {0.3, 0.4, 0.5, 0.6}), the decision rule and ±2-frame scene matching are unchanged. Seed 42 is the main-study run.

Part	Content
Part 1	Extra seeds for A0 and A1 (training)
Part 2	Protocol A with ℓ_min = 1 on all test sets (evaluation only)
Part 3	Protocol B, validation-selected thresholds (evaluation only)
Part 4	Additional check: validation F1 for every epoch

Part 1: Extra seeds for A0 and A1 (training)

A0 (full model, dual-label) and A1 (full model, midpoint only) were trained with seeds 43 and 44, giving four new runs (A0-s43, A0-s44, A1-s43, A1-s44). Recipe, augmentation, synthetic transitions, EMA (0.999), early stopping (patience 10 on SHOT validation F1 with the variant's own decision rule) and the 80-epoch maximum are identical to the seed-42 runs. Only the training seed changes: model initialisation, training-data order, the ClipShots clip drawn each epoch, augmentation and synthetic transitions.

1.1 Train/validation split is identical for all seeds

In the training code the SHOT and ClipShots train/validation split was shuffled with the training seed, so seeds 43 and 44 would have used different validation videos. I fixed the split to the seed-42 split for every training seed (the option defaults to 42, so the seed-42 runs are unchanged). SHA-256 of the sorted SHOT validation video names:

Training seed	Old rule (split follows the seed)	Fixed split (used)
42	f0070e4a…	f0070e4a…
43	88ddfd45…	f0070e4a…
44	792acaf5…	f0070e4a…

All four new runs printed the same checksums before training: SHOT validation 67 videos (f0070e4a…), SHOT train 381 videos (ed9cb6ec…), ClipShots train 3,108 videos (f623f785…). The pre-extracted training clips use the stable per-video MD5 seed and are unchanged.

1.2 The four new runs

Run	Seed	Best epoch	Best SHOT val F1	Epochs run	Stopped early	Training hours
A0	42	21	0.8029	31	True	5.40
A0-s43	43	13	0.7953	23	True	6.97
A0-s44	44	16	0.7935	26	True	7.26
A1	42	7	0.7863	17	True	2.92
A1-s43	43	10	0.7915	20	True	6.46
A1-s44	44	9	0.7967	19	True	6.18

Seed-42 rows are the main-study runs, shown for reference.

1.3 SHOT validation F1 history of every epoch

SHOT validation F1 after every epoch; best epoch in bold. Seed 42 runs are A1 and A0.

Epoch	A1	A0	A1-s43	A1-s44	A0-s43	A0-s44
1	0.7029	0.6988	0.7257	0.7280	0.7223	0.7087
2	0.7499	0.7514	0.7659	0.7515	0.7586	0.7446
3	0.7535	0.7543	0.7715	0.7664	0.7615	0.7487
4	0.7628	0.7774	0.7845	0.7902	0.7893	0.7827
5	0.7807	0.7758	0.7891	0.7910	0.7881	0.7885
6	0.7796	0.7722	0.7877	0.7909	0.7850	0.7888
7	**0.7863**	0.7725	0.7891	0.7940	0.7862	0.7867
8	0.7785	0.7748	0.7863	0.7927	0.7902	0.7880
9	0.7695	0.7654	0.7909	**0.7967**	0.7801	0.7782

Epoch	A1	A0	A1-s43	A1-s44	A0-s43	A0-s44
10	0.7852	0.7838	**0.7915**	0.7949	0.7929	0.7857
11	0.7843	0.7793	0.7822	0.7935	0.7835	0.7801
12	0.7783	0.7843	0.7818	0.7864	0.7889	0.7818
13	0.7824	0.7821	0.7901	0.7812	**0.7953**	0.7706
14	0.7704	0.7681	0.7835	0.7941	0.7853	0.7825
15	0.7840	0.7885	0.7865	0.7859	0.7930	0.7847
16	0.7735	0.7926	0.7782	0.7909	0.7864	**0.7935**
17	0.7846	0.7810	0.7774	0.7874	0.7781	0.7888
18		0.7932	0.7882	0.7914	0.7858	0.7912
19		0.7946	0.7733	0.7898	0.7806	0.7825
20		0.7960	0.7830		0.7926	0.7866
21		**0.8029**			0.7832	0.7828
22		0.7935			0.7883	0.7882
23		0.7907			0.7933	0.7908
24		0.7953				0.7870
25		0.7854				0.7820
26		0.7847				0.7850
27		0.7913				
28		0.7951				
29		0.7810				
30		0.7886				
31		0.7870				

1.4 Does training several runs at once on one GPU change the results?

A seed-42 A1 run trained alongside the four new runs (five jobs on one GPU) was compared with the original seed-42 A1 run over its first two epochs. Train loss 0.03501 vs 0.03502 (epoch 1) and 0.01847 vs 0.01849 (epoch 2); SHOT validation F1 0.7046 vs 0.7029 (epoch 1) and 0.7505 vs 0.7499 (epoch 2). The two runs are **not bit-identical**, so a repeated training run, with or without concurrent jobs, gives slightly different numbers; I could not separate concurrency from other run-to-run non-determinism. The differences are small (at most 0.002 validation F1 and 0.0001 train loss) and far below the spread between seeds at epoch 2 (validation F1 0.7446–0.7659 across the four new runs). Data-loader seeding is per run and the split is fixed, so concurrency does not change the data each run sees.

Part 2: Protocol A with ℓ_min = 1 (evaluation only)

ℓ_min is fixed to 1 (no minimum-shot-length merging) on SHOT, ClipShots and BBC and is not searched. Thresholds are still selected on each test set (Protocol A).

2.1 Checks

The evaluation pipeline was checked against the earlier evaluation of the same checkpoints and reproduces it exactly for all 17 seed-42 models (51 of 51 model-test-set results: F1, thresholds, TP / FP / FN).

Saved probabilities vs a fresh forward pass (A0 and A1, all three test sets): the maximum absolute difference of p_y and p_z over all videos is 0.0, and F1, thresholds and TP / FP / FN are identical. The results in Parts 2 and 3 therefore use the saved probabilities.

Results

2.2 Supervision (Table 7a, seed 42)

ID	Description	SHOT F1	ClipShots F1	BBC F1
A0	Full, dual-label (primary)	0.8449	0.8001	0.9719
A1	Full, midpoint only	0.8478	0.7609	0.9698
A2	Full, unified single head	**0.8509**	0.7709	0.9473
A3	Reduced, midpoint only	0.8312	0.7606	0.9714
A4	Reduced, dual-label	0.8370	**0.8045**	**0.9721**

Differences:

Delta	SHOT	ClipShots	BBC
A0 – A1 (full)	-0.0029	+0.0392	+0.0021
A4 – A3 (reduced)	+0.0058	+0.0439	+0.0007
A0 – A2 (dual vs unified)	-0.0060	+0.0292	+0.0245

2.3 Components, leave-one-out (Table 7b)

ID	Description	SHOT F1	ClipShots F1	BBC F1
A0	Full model	0.8449	0.8001	0.9719
C1	w/o difference-based branch (B)	**0.8536**	0.7942	0.9701
C2a	w/o FrameSimilarity	0.8457	0.8009	0.9681
C2b	w/o ColorHistogramSimilarity	0.8361	**0.8077**	0.9730
C3	w/o temporal delta	0.8363	0.7956	0.9728
C4	w/o per-frame pathway (relations from Branch A)	0.8420	0.8068	0.9699
C5a	w/o pixel-level motion descriptor	0.8487	0.8021	0.9715
C5b	w/o edge-density head	0.8454	0.7967	**0.9733**
C6	w/o self-attention and gate	0.8364	0.7969	0.9667

Change in F1 relative to A0:

ID	ΔSHOT	ΔClipShots	ΔBBC
C1	+0.0087	-0.0059	-0.0017
C2a	+0.0008	+0.0008	-0.0037
C2b	-0.0089	+0.0076	+0.0012
C3	-0.0086	-0.0045	+0.0009
C4	-0.0029	+0.0067	-0.0020
C5a	+0.0037	+0.0020	-0.0003
C5b	+0.0004	-0.0034	+0.0015
C6	-0.0085	-0.0032	-0.0052

Largest single-component change on any dataset: 0.0089.

2.4 Build-up ladder

Step	Run	What is added	SHOT F1	ClipShots F1	BBC F1
0	M0_min	RGB encoder + MLP + dual head only	0.8169	0.7555	0.9543
1	M1_edge	+ edge-density head	0.8261	0.7672	0.9584
2	B_noatt	+ 4 fusion blocks, no attention	0.8351	0.7984	0.9682
3	A4	+ attention and gate (reduced model)	0.8370	0.8045	**0.9721**
4	S1_rel	+ FrameSimilarity, ColorHist, delta	0.8392	**0.8054**	0.9706
5	A0	+ diff branch, motion, per-frame pathway (full model)	**0.8449**	0.8001	0.9719

Change at each step:

Step	SHOT	ClipShots	BBC
M0_min → M1_edge	+0.0091	+0.0117	+0.0041
M1_edge → B_noatt	+0.0091	+0.0312	+0.0098
B_noatt → A4	+0.0018	+0.0061	+0.0039
A4 → S1_rel	+0.0023	+0.0008	-0.0015
S1_rel → A0	+0.0057	-0.0053	+0.0013

2.5 Precision, recall and selected thresholds

ID	SHOT P / R	ClipShots P / R	BBC P / R	SHOT τ_c, τ_g	ClipShots τ_c, τ_g	BBC τ_c, τ_g
A0	0.842 / 0.848	0.773 / 0.829	0.963 / 0.980	0.4, 0.3	0.65, 0.6	0.45, 0.6
A1	0.858 / 0.838	0.702 / 0.830	0.975 / 0.964	0.45, –	0.45, –	0.45, –
A2	0.856 / 0.846	0.741 / 0.803	0.926 / 0.969	0.45, –	0.65, –	0.5, –
A3	0.869 / 0.797	0.701 / 0.832	0.966 / 0.977	0.45, –	0.45, –	0.45, –
A4	0.839 / 0.835	0.786 / 0.824	0.971 / 0.973	0.4, 0.4	0.7, 0.6	0.5, 0.6
C1	0.864 / 0.844	0.775 / 0.815	0.968 / 0.972	0.4, 0.3	0.7, 0.5	0.45, 0.6
C2a	0.854 / 0.837	0.776 / 0.828	0.970 / 0.966	0.45, 0.5	0.65, 0.6	0.5, 0.6
C2b	0.839 / 0.833	0.776 / 0.842	0.967 / 0.979	0.3, 0.3	0.65, 0.6	0.45, 0.6
C3	0.813 / 0.861	0.758 / 0.838	0.961 / 0.985	0.4, 0.4	0.65, 0.6	0.45, 0.6
C4	0.857 / 0.827	0.792 / 0.822	0.957 / 0.983	0.5, 0.3	0.7, 0.6	0.45, 0.6
C5a	0.841 / 0.857	0.772 / 0.834	0.960 / 0.983	0.4, 0.3	0.65, 0.6	0.45, 0.6
C5b	0.839 / 0.852	0.762 / 0.835	0.964 / 0.983	0.35, 0.4	0.65, 0.6	0.45, 0.6
C6	0.839 / 0.834	0.773 / 0.823	0.954 / 0.979	0.45, 0.4	0.7, 0.6	0.45, 0.6
M0_min	0.795 / 0.840	0.745 / 0.766	0.957 / 0.952	0.45, 0.5	0.6, 0.6	0.45, 0.6
M1_edge	0.839 / 0.814	0.741 / 0.795	0.964 / 0.953	0.4, 0.6	0.55, 0.6	0.45, 0.6
B_noatt	0.845 / 0.826	0.773 / 0.826	0.960 / 0.976	0.45, 0.5	0.7, 0.6	0.45, 0.6

ID	SHOT P / R	ClipShots P / R	BBC P / R	SHOT τ_c, τ_g	ClipShots τ_c, τ_g	BBC τ_c, τ_g
S1_rel	0.846 / 0.833	0.782 / 0.830	0.960 / 0.981	0.4, 0.4	0.65, 0.5	0.45, 0.6
A0-s43	0.881 / 0.814	0.765 / 0.825	0.960 / 0.971	0.4, 0.4	0.55, 0.6	0.4, 0.6
A0-s44	0.853 / 0.844	0.800 / 0.807	0.964 / 0.977	0.4, 0.4	0.65, 0.6	0.45, 0.6
A1-s43	0.863 / 0.830	0.702 / 0.822	0.964 / 0.970	0.45, –	0.45, –	0.4, –
A1-s44	0.850 / 0.837	0.712 / 0.797	0.967 / 0.971	0.45, –	0.5, –	0.45, –

A1, A2, A3 and their seeds use τ_c only.

2.6 Table 8 — Per-type analysis on ClipShots

ID	Cut recall	Gradual recall	Gradual F1	Gradual IoU
A0	0.8484	0.7866	0.7415	0.6232
A1	0.9093	0.6623	n/a	n/a
A2	0.8011	0.8070	n/a	0.6965

2.7 Table 9 — Recall by gradual-transition duration on ClipShots

ID	≤ 10 frames	11–30 frames	> 30 frames
A0	0.7740 (n = 832)	0.8018 (n = 1241)	0.7500 (n = 228)
C6	0.7500 (n = 832)	0.8251 (n = 1241)	0.7368 (n = 228)

C6 – A0: ≤ 10 -0.0240, 11–30 +0.0234, > 30 -0.0132.

2.8 Seeds table

F1 of A0 and A1 at seeds 42, 43 and 44 (seed 42 = the main-study run). Mean and sample standard deviation (n – 1) over the three seeds.

Model	Test set	Seed 42	Seed 43	Seed 44	Mean	SD
A0	SHOT	0.8449	0.8463	0.8486	0.8466	0.0018
A0	ClipShots	0.8001	0.7938	0.8031	0.7990	0.0047
A0	BBC	0.9719	0.9655	0.9706	0.9693	0.0034
A1	SHOT	0.8478	0.8460	0.8436	0.8458	0.0021
A1	ClipShots	0.7609	0.7573	0.7524	0.7569	0.0043
A1	BBC	0.9698	0.9670	0.9691	0.9686	0.0015

Paired difference A0 – A1 (same seed for both):

Test set	Seed 42	Seed 43	Seed 44	Mean	SD
SHOT	-0.0029	+0.0003	+0.0049	+0.0008	0.0039
ClipShots	+0.0392	+0.0366	+0.0507	+0.0421	0.0075
BBC	+0.0021	-0.0014	+0.0015	+0.0007	0.0019

Cut recall and gradual recall on ClipShots:

Model	Metric	Seed 42	Seed 43	Seed 44	Mean	SD
A0	Cut recall	0.8484	0.8423	0.8341	0.8416	0.0072
A0	Gradual recall	0.7866	0.7870	0.7479	0.7739	0.0225
A1	Cut recall	0.9093	0.9048	0.8914	0.9019	0.0093
A1	Gradual recall	0.6623	0.6454	0.5958	0.6345	0.0346

2.9 ℓ_min sensitivity for A0 on ClipShots

Best F1 for each fixed ℓ_min, thresholds re-searched for each value:

ℓ_min	τ_c	τ_g	Precision	Recall	F1
1	0.65	0.6	0.7734	0.8287	0.8001
2	0.65	0.6	0.7800	0.8274	0.8030
3	0.65	0.6	0.7848	0.8256	0.8047
4	0.65	0.6	0.7871	0.8241	**0.8052**

2.10 Interpretation check (0.01 band)

Statement	Value	Holds?
A0 – A1 on ClipShots ≥ 0.01, seed 42	+0.0392	**yes**
A0 – A1 on ClipShots ≥ 0.01, mean over the three seeds	+0.0421	**yes**
A0 – A1 on ClipShots positive for every seed	+0.0392 / +0.0366 / +0.0507	**yes**
A4 – A3 on ClipShots ≥ 0.01	+0.0439	**yes**
A0 – A2 on ClipShots ≥ 0.01	+0.0292	**yes**
A0 – A2 on BBC ≥ 0.01	+0.0245	**yes**
No leave-one-out change (C1–C6) reaches 0.01 on any test set	largest 0.0089	**yes**

Part 3: Protocol B, validation-selected thresholds (evaluation only)

For each model τ_c and τ_g are selected once on the SHOT validation videos and applied unchanged to SHOT, ClipShots and BBC ($\ell_{min} = 1$). Nothing is tuned on a test set. Midpoint-only (A1, A3) and unified (A2) models select τ_c only.

3.1 Validation videos as full videos (prerequisite)

Before this work the 67 SHOT validation videos existed only as pre-cut training clips, so I **added a data path**: the 67 videos are now decoded completely with the same decoder, resizing (48 × 64), window layout (T = 128, stride 64) and window averaging as the SHOT test set (1,270 windows), and scored with the same scene matching. Ground truth comes from the SHOT label file with the same transition-to-scene conversion as ClipShots; 720 of 775 transitions lie within the decoded video length and are used. The label file is the only ground truth available for these videos and is not the authors' final annotation, so validation F1 is noisier than test F1 and the selected thresholds are noisier too.

3.2 Threshold selection and application

$\tau_c \in \{0.10, \ldots, 0.90\}$ and $\tau_g \in \{0.3, 0.4, 0.5, 0.6\}$ were searched once per model by maximising F1 on the full validation videos with $\ell_{min} = 1$. The selected values and the validation F1 are in the thresholds table (3.6). They were applied unchanged to the three test sets for every model, including the new seeds. Numbers are expected to be lower than under Protocol A, especially on ClipShots and BBC.

Results

3.3 Supervision (Table 7a, seed 42)

ID	Description	SHOT F1	ClipShots F1	BBC F1
A0	Full, dual-label (primary)	0.8084	**0.7658**	0.9439
A1	Full, midpoint only	0.8313	0.7352	0.9597
A2	Full, unified single head	0.8350	0.7589	0.9473
A3	Reduced, midpoint only	0.8223	0.7562	**0.9713**
A4	Reduced, dual-label	**0.8364**	0.7627	0.9660

Differences:

Delta	SHOT	ClipShots	BBC
A0 – A1 (full)	-0.0230	+0.0305	-0.0158
A4 – A3 (reduced)	+0.0141	+0.0065	-0.0054
A0 – A2 (dual vs unified)	-0.0267	+0.0069	-0.0034

3.4 Components, leave-one-out (Table 7b)

ID	Description	SHOT F1	ClipShots F1	BBC F1
A0	Full model	0.8084	0.7658	0.9439
C1	w/o difference-based branch (B)	0.8357	0.7758	0.9444
C2a	w/o FrameSimilarity	**0.8408**	0.7265	0.9449
C2b	w/o ColorHistogramSimilarity	0.8260	0.7513	0.9546
C3	w/o temporal delta	0.8208	0.7709	0.9583
C4	w/o per-frame pathway (relations from Branch A)	0.8310	0.7920	0.9605
C5a	w/o pixel-level motion descriptor	0.8154	**0.7951**	0.9663
C5b	w/o edge-density head	0.8340	0.7638	**0.9680**
C6	w/o self-attention and gate	0.8147	0.7708	0.9534

Change in F1 relative to A0:

ID	ΔSHOT	ΔClipShots	ΔBBC
C1	+0.0274	+0.0101	+0.0005
C2a	+0.0325	-0.0393	+0.0009
C2b	+0.0176	-0.0144	+0.0107
C3	+0.0124	+0.0052	+0.0144
C4	+0.0226	+0.0262	+0.0165
C5a	+0.0070	+0.0294	+0.0224
C5b	+0.0257	-0.0020	+0.0241
C6	+0.0063	+0.0050	+0.0095

Largest single-component change on any dataset: 0.0393.

3.5 Build-up ladder

Step	Run	What is added	SHOT F1	ClipShots F1	BBC F1
0	M0_min	RGB encoder + MLP + dual head only	0.8124	0.7396	0.9502
1	M1_edge	+ edge-density head	0.8082	0.7633	0.9513
2	B_noatt	+ 4 fusion blocks, no attention	0.8259	0.7791	0.9525
3	A4	+ attention and gate (reduced model)	**0.8364**	0.7627	**0.9660**
4	S1_rel	+ FrameSimilarity, ColorHist, delta	0.8275	**0.7862**	0.9534
5	A0	+ diff branch, motion, per-frame pathway (full model)	0.8084	0.7658	0.9439

Change at each step:

Step	SHOT	ClipShots	BBC
M0_min → M1_edge	-0.0042	+0.0236	+0.0011
M1_edge → B_noatt	+0.0177	+0.0159	+0.0012
B_noatt → A4	+0.0105	-0.0164	+0.0135
A4 → S1_rel	-0.0089	+0.0234	-0.0125
S1_rel → A0	-0.0191	-0.0204	-0.0095

3.6 Precision, recall and selected thresholds

ID	τ_c, τ_g (from validation)	Validation F1	SHOT P / R	ClipShots P / R	BBC P / R
A0	0.6, 0.4	0.7749	0.883 / 0.745	0.684 / 0.869	0.929 / 0.959
A1	0.55, –	0.7482	0.909 / 0.766	0.744 / 0.726	0.983 / 0.938

ID	τ_c, τ_g (from validation)	Validation F1	SHOT P / R	ClipShots P / R	BBC P / R
A2	0.55, –	0.7583	0.892 / 0.785	0.679 / 0.859	0.939 / 0.956
A3	0.5, –	0.7413	0.888 / 0.766	0.730 / 0.784	0.977 / 0.966
A4	0.45, 0.5	0.7405	0.865 / 0.810	0.665 / 0.894	0.948 / 0.985
C1	0.55, 0.3	0.7712	0.899 / 0.781	0.700 / 0.870	0.933 / 0.956
C2a	0.5, 0.4	0.7618	0.863 / 0.820	0.613 / 0.891	0.917 / 0.974
C2b	0.45, 0.3	0.7632	0.878 / 0.780	0.648 / 0.893	0.927 / 0.983
C3	0.55, 0.5	0.7588	0.863 / 0.783	0.685 / 0.882	0.950 / 0.966
C4	0.55, 0.5	0.7664	0.887 / 0.782	0.723 / 0.876	0.953 / 0.968
C5a	0.55, 0.6	0.7581	0.898 / 0.747	0.739 / 0.861	0.968 / 0.964
C5b	0.45, 0.5	0.7672	0.863 / 0.807	0.665 / 0.898	0.952 / 0.985
C6	0.55, 0.5	0.7539	0.862 / 0.772	0.686 / 0.880	0.946 / 0.961
M0_min	0.5, 0.6	0.7235	0.834 / 0.792	0.686 / 0.803	0.972 / 0.930
M1_edge	0.5, 0.6	0.7363	0.879 / 0.748	0.720 / 0.813	0.973 / 0.931
B_noatt	0.55, 0.5	0.7387	0.872 / 0.784	0.704 / 0.872	0.949 / 0.956
S1_rel	0.5, 0.4	0.7598	0.894 / 0.770	0.717 / 0.871	0.931 / 0.977
A0-s43	0.5, 0.4	0.7751	0.907 / 0.754	0.677 / 0.863	0.929 / 0.948
A0-s44	0.45, 0.4	0.7625	0.870 / 0.828	0.675 / 0.883	0.924 / 0.981
A1-s43	0.55, –	0.7588	0.909 / 0.764	0.763 / 0.711	0.984 / 0.926
A1-s44	0.55, –	0.7602	0.902 / 0.757	0.736 / 0.746	0.981 / 0.939

A1, A2, A3 and their seeds use τ_c only.

3.7 Table 8 — Per-type analysis on ClipShots

ID	Cut recall	Gradual recall	Gradual F1	Gradual IoU
A0	0.8712	0.8640	0.6488	0.7338
A1	0.8810	0.3968	n/a	n/a
A2	0.8718	0.8331	n/a	0.7218

3.8 Table 9 — Recall by gradual-transition duration on ClipShots

ID	≤ 10 frames	11–30 frames	> 30 frames
A0	0.8341 (n = 832)	0.8759 (n = 1241)	0.9079 (n = 228)
C6	0.8257 (n = 832)	0.8767 (n = 1241)	0.8421 (n = 228)

C6 – A0: ≤ 10 -0.0084, 11–30 +0.0008, > 30 -0.0658.

3.9 Seeds table

F1 of A0 and A1 at seeds 42, 43 and 44 (seed 42 = the main-study run). Mean and sample standard deviation (n – 1) over the three seeds.

Model	Test set	Seed 42	Seed 43	Seed 44	Mean	SD
A0	SHOT	0.8084	0.8236	0.8483	0.8268	0.0202
A0	ClipShots	0.7658	0.7588	0.7653	0.7633	0.0039
A0	BBC	0.9439	0.9383	0.9514	0.9446	0.0066
A1	SHOT	0.8313	0.8303	0.8228	0.8281	0.0046
A1	ClipShots	0.7352	0.7360	0.7410	0.7374	0.0032
A1	BBC	0.9597	0.9545	0.9594	0.9579	0.0029

Paired difference A0 – A1 (same seed for both):

Test set	Seed 42	Seed 43	Seed 44	Mean	SD
SHOT	-0.0230	-0.0067	+0.0255	-0.0014	0.0247
ClipShots	+0.0305	+0.0228	+0.0242	+0.0259	0.0041
BBC	-0.0158	-0.0162	-0.0080	-0.0133	0.0046

Cut recall and gradual recall on ClipShots:

Model	Metric	Seed 42	Seed 43	Seed 44	Mean	SD
A0	Cut recall	0.8712	0.8663	0.9067	0.8814	0.0220
A0	Gradual recall	0.8640	0.8561	0.8340	0.8514	0.0156
A1	Cut recall	0.8810	0.8725	0.8775	0.8770	0.0043
A1	Gradual recall	0.3968	0.3659	0.4654	0.4094	0.0509

3.10 Interpretation check (0.01 band)

Statement	Value	Holds?
A0 – A1 on ClipShots ≥ 0.01, seed 42	+0.0305	**yes**
A0 – A1 on ClipShots ≥ 0.01, mean over the three seeds	+0.0259	**yes**
A0 – A1 on ClipShots positive for every seed	+0.0305 / +0.0228 / +0.0242	**yes**
A4 – A3 on ClipShots ≥ 0.01	+0.0065	**no**
A0 – A2 on ClipShots ≥ 0.01	+0.0069	**no**
A0 – A2 on BBC ≥ 0.01	-0.0034	**no**
No leave-one-out change (C1–C6) reaches 0.01 on any test set	largest 0.0393	**no**

Differences below 0.01 F1 are treated as inconclusive. Under Protocol B the thresholds themselves add noise, so single-model differences are less reliable than under Protocol A.

Part 4: Additional check — SHOT validation F1 for every epoch

The validation F1 of every epoch for A0 and A1 at seed 42 and for the four new runs is in Table 1.3. What happens after each run's best epoch:

4.1 Behaviour after the best epoch

Run	Best epoch	Best val F1	Stopped at epoch	Range after best	Last	Trend per epoch
A1	7	0.7863	17	0.7695–0.7852	0.7846	+0.00024
A0	21	0.8029	31	0.7810–0.7953	0.7870	-0.00066
A1-s43	10	0.7915	20	0.7733–0.7901	0.7830	-0.00053
A1-s44	9	0.7967	19	0.7812–0.7949	0.7898	-0.00021
A0-s43	13	0.7953	23	0.7781–0.7933	0.7933	+0.00037
A0-s44	16	0.7935	26	0.7820–0.7912	0.7850	-0.00036

4.2 Did validation F1 level off or fall after the best epoch?

In every one of the six runs, including A1 at seed 42, early stopping ended training after 10 epochs without improvement, and validation F1 had **levelled off** rather than still rising or collapsing: after its best epoch each run moves up and down inside a band about 0.01–0.02 wide just below its best value, with a linear trend close to zero (between –0.0007 and +0.0004 F1 per epoch), and no run was still climbing when it stopped. A1 at seed 42 (best epoch 7, 0.7863) fluctuated between 0.7695 and 0.7852 afterwards and ended at 0.7846, so it was levelled off and unstable at the 0.01 level, not falling steadily. The new A1 runs behave the same way (A1-s43 drifts slightly down, –0.0005 per epoch; A1-s44 is flat, –0.0002). The epoch-to-epoch swings are about as large as the gaps between runs, so the best-epoch choice itself is noisy.